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CSA06 – Design and Analysis of Algorithms

CO1 – Assessment Tool 1: IT Application-Based Problem Solving

Scalability Audit of an E-Commerce Platform

Course Code & Name	CSA06 – Design and Analysis of Algorithms
Assessment	CO1 – Assessment Tool 1: IT Application-Based Problem Solving – Scalability Audit of an E-Commerce Platform
Course Outcome Assessed	Determine algorithm efficiency using asymptotic notations and mathematical techniques including the Master Theorem and the substitution method
Duration	3 Days
Mode	Individual
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Instructions: Answer all parts. Show the basic operation, every step of each derivation and the final asymptotic result, and interpret what each result means for the platform. Assume the platform's servers execute 10^[exponent missing in the original paper] basic operations per second.

Scenario: Scalability Audit of "ShopSphere", an Online Shopping Platform

ShopSphere is a growing e-commerce platform. Its product catalogue currently holds **n = 100,000** products and is expected to reach **n = 1,000,000** within a year. Before the expansion, the CTO has asked you, as the platform's algorithm engineer, to audit five back-end modules, predict how each will behave as n grows, and recommend which modules must be redesigned.

Module M1: Duplicate-Listing Detector

Sellers sometimes upload the same product twice. Every night, this routine compares every product with every later product.

```
FOR i ← 0 TO n − 2 DO
  FOR j ← i + 1 TO n − 1 DO
    IF Product[i].title = Product[j].title AND
       Product[i].seller = Product[j].seller THEN
       MarkDuplicate(j)
    END IF
  END FOR
END FOR
```

Module M2: API Rate Limiter

The public API audits request windows of doubling size using the routine below. A team member claims it is Θ(n log n) because the outer loop runs about log n times and the inner loop runs up to n times.

```
i ← 1
WHILE i < n DO
  FOR j ← 0 TO i − 1 DO
    Audit(Request[j])        // basic operation
  END FOR
  i ← i × 2
END WHILE
```

Module M3: Product Search

Two versions exist.

- **S1** scans the unsorted catalogue sequentially until the product is found.
- **S2** searches a sorted product index by comparing the key with the middle entry and discarding half the index after each comparison. Its comparison count satisfies T(n) = T(n/2) + 1, T(1) = 1.

Module M4: Recommendation Engine

The engine splits the customer–product data into four parts of size n/2 (because of replicated partitions) and merges the partial results in linear time: $\mathbf{T(n) = 4T(n/2) + n}$. A proposed redesign removes the replication, giving $\mathbf{T(n) = 2T(n/2) + n}$.

Module M5: Sales Analytics Engine

The engine splits the sales data into halves recursively and sorts the records by revenue at every level, giving $\mathbf{T(n) = 2T(n/2) + n \log n}$. Separately, profiling shows that the nightly report generator performs $\mathbf{f(n) = 3n^2 + 10n \log n + 50}$ operations.

Questions

(a) Understand the Problem. For each of the five modules, identify the input size and the basic operation that dominates the running time, explain where it occurs, and state how you expect its execution count to grow as n increases. Justify each choice briefly.

(b) Analyse the Iterative Modules. For M1, derive the exact number of comparisons as a function of n and state the time and space complexity. For M2, derive the exact number of Audit operations, decide whether the team member's $\Theta(n \log n)$ claim is correct, and give the tight bound. Verify your M2 result for n = 1,024.

(c) Solve a Recurrence by Substitution. Solve the recurrence for search version S2 using backward substitution, showing every intermediate step, and derive the exact closed form. State the time and space complexity of both an iterative and a recursive implementation.

(d) Apply the Master Theorem. For M4, apply the Master Theorem to both the current design and the proposed redesign, clearly identifying a, b, f(n), n^(log_b a) and the case used, and explain what happens to the running time when the data doubles. For M5, determine whether any of the three standard cases applies to T(n) = 2T(n/2) + n log n; if not, derive the complexity using the extended Master Theorem or a recursion tree.

(e) Analyse Best, Average and Worst Cases. For search versions S1 and S2, identify the input condition that produces the best, average and worst case and give the number of comparisons in each. For S1, derive the average number of comparisons when the probability that the searched product exists is p, and evaluate it for n = 1,000,000 and p = 0.5.

(f) Prove an Asymptotic Bound. Using the formal definitions, prove that the report generator's cost $f(n) = 3n^2 + 10n \log n + 50$ is $\Theta(n^2)$ by finding constants c_1, c_2 and n_0, and show that $f(n)$ is not $O(n)$.

(g) Scalability Decision. Estimate the number of basic operations and the running time of M1, M2, S2, M4 (both designs) and M5 for n = 1,000,000. Identify the modules that will fail to meet a nightly processing window of 10 minutes, and recommend, with justification, which modules must be redesigned and how.

Rubrics

Criteria	**Weightage (%)**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured and	Mostly logical with minor gaps	Basic approach with some	Illogical or unclear approach

		efficient approach		inconsistencies	
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized and properly explained	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Code of Conduct: I _____ (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Date: _____